

Question 2:

Compute the average power, in units of watts, the object requires to enable this main use case. Clearly demonstrate your work, starting from first principles. List the assumptions, theory, equations and/or method used to compute an average value in watts.

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Sienna's Response Q2

The objective of this analysis was to determine the average electrical power required for my Microsoft Surface Laptop Studio 2 to perform its primary use case: web browsing and online research. Rather than using the laptop charger's maximum power rating, I experimentally measured energy consumption during a representative browsing session. I first calculated average power from the change in stored battery energy and then used HWiNFO64 to independently record instantaneous battery discharge power.

Power is the rate at which energy is transferred or consumed. Therefore, average power can be calculated from first principles as

$$P_{avg}=\Delta E/\Delta t$$

where P_{avg} is average power in watts (W), ΔE is energy consumed in watt-hours (Wh), and Δt is elapsed time in hours (h). Because the battery stores energy in watt-hours, dividing the decrease in stored battery energy by time directly produces average power in watts. This method measures the total system power required by the laptop, including the processor, memory, storage, display, Wi-Fi hardware, motherboard, keyboard, and cooling system.

To approximate normal web browsing while controlling major variables, the laptop began at 100% charge and remained unplugged throughout the experiment. Screen brightness was fixed at 50%, Wi-Fi was enabled, Bluetooth was disabled, Battery Saver was disabled, and no external monitors or peripherals were connected. Google Chrome was used to conduct searches, navigate webpages, read articles, and take notes. Gaming, video streaming, software installations, and other intentionally computationally intensive activities were avoided. I assumed that the reported battery state of charge was proportional to the battery's available stored energy and that background system activity represented normal laptop operation.

Using Windows' `powercfg/batteryreport` command, I determined that the battery had a full-charge capacity of approximately **56.85 Wh**. During a 30-minute browsing experiment, the displayed battery charge decreased by **13 percentage points**, from 100% to 87%. The estimated electrical energy consumed was therefore

$$\Delta E=(0.13)(56.85\text{ Wh})=7.39\text{ Wh}$$

The elapsed time was $\Delta t=0.5$ hours

Therefore, $P_{avg}=(0.50h)(7.39\text{ Wh})=14.78\text{ W}$

This first-principles calculation estimates that the laptop required an average of approximately **14.8 W** during the defined browsing workload.

To independently evaluate this estimate, I performed a second test using HWiNFO64, which continuously logged the battery charge/discharge rate rather than relying on only the initial and final displayed battery percentages. The recorded power fluctuated throughout the session as webpages loaded, processor activity changed, and periods of reading required less computation. The logged dataset produced an overall average discharge magnitude of approximately **10.17 W**. During this test, the laptop also shut off and turned back on due to brief perceived inactivity; this interruption is visible in the power-versus-time data and could contribute to the lower average result.

The HWiNFO measurement is lower than the 14.78 W estimate from the battery-percentage method. This difference highlights an important limitation of the first method: Windows displays battery percentage in discrete increments, so measuring a relatively small change over only 30 minutes introduces substantial uncertainty. The browsing workload also varies naturally with user activity and background processes. Continuous power logging therefore provides a higher-resolution characterization of instantaneous system behavior, while the battery-energy calculation demonstrates the average-power calculation directly from first principles.

Together, the experiments indicate that web browsing on this laptop requires power on the order of **10–15 W**, with **14.78 W** as the first-principles estimate. These measurements establish the baseline against which the proposed redesign can be evaluated under an identical workload.

Windows Battery Report

Battery report

COMPUTER NAME	SIENNASURFACE
SYSTEM PRODUCT NAME	Microsoft Corporation Surface Laptop Studio 2
BIOS	23.101.143 11/27/2025
OS BUILD	26100.1.amd64fre.ge_release.240331-1435
PLATFORM ROLE	Mobile
CONNECTED STANDBY	Supported
REPORT TIME	2026-08-11 00:02:00

Installed batteries

Information about each currently installed battery

BATTERY 1

NAME	SurfaceBattery
MANUFACTURER	SMP
SERIAL NUMBER	2004107519
CHEMISTRY	LION
DESIGN CAPACITY	56,310 mWh
FULL CHARGE CAPACITY	56,770 mWh
CYCLE COUNT	96

HWiNFO Power Consumption Graph

